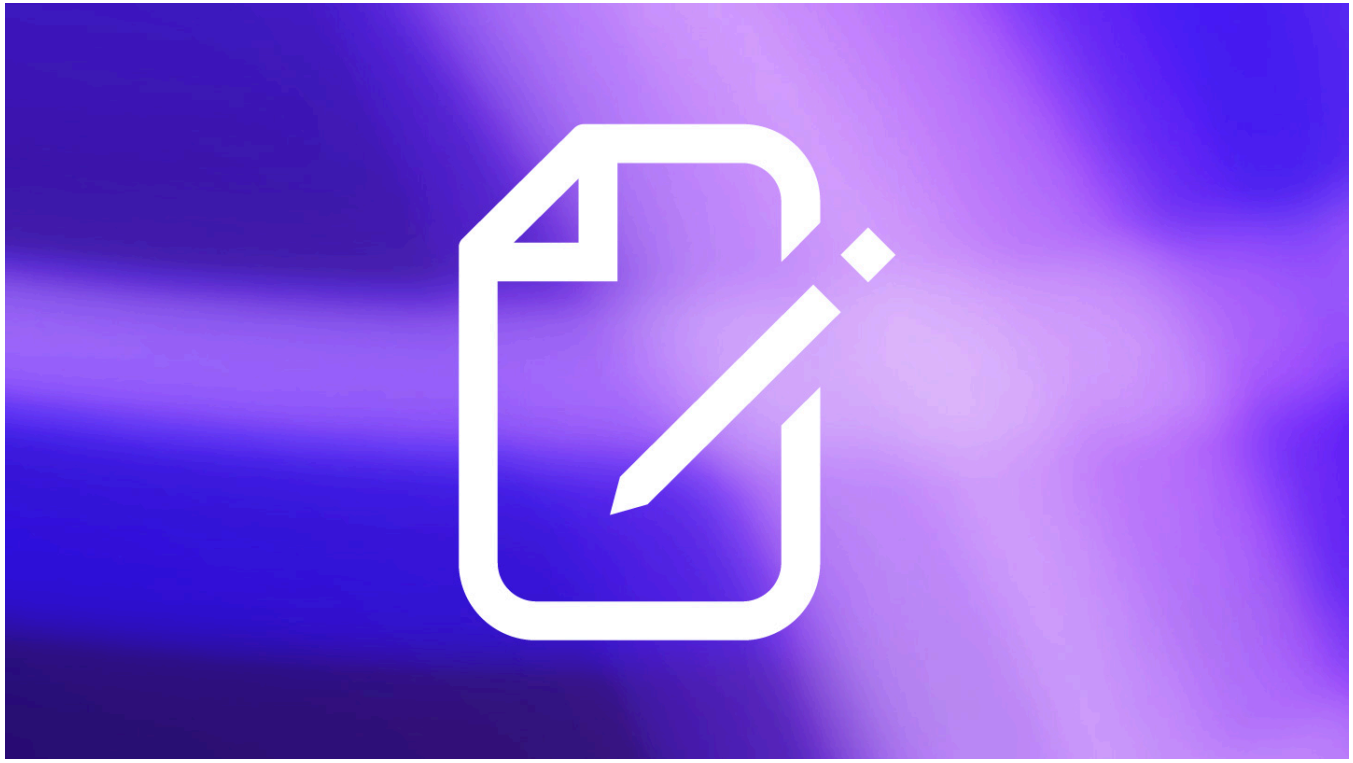
National

Install and troubleshoot the Ethernet Passive Optical Network (EPON) two-box (Optical Node Unit (ONU) and XB7 or XER10) set-up for residential

HOW21267 | MODIFIED 06/10/2026



Summary

The article provides a comprehensive technical overview and installation guide for Comcast's enhanced Fiber to the Premise (FTTP) service using an EPON with a two-box device solution (ONU, plus Technicolor XB7 or Sercomm XER10 router) and EPON with customer-owned and managed equipment (COAM). It aims to inform technicians about device configurations, installation procedures, troubleshooting, and service activation to improve deployment efficiency and customer experience for Xfinity TV, Internet, Home, and Voice services.

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What's changed in this article

Date	Summary of change
June 10, 2026	Added minor text revisions and one replacement photo.
June 3, 2026	Article reviewed and refreshed.
April 6, 2026	Adding in XF-ONU2.
December 15, 2025	Minor formatting adjustments.
November 25, 2025	Revised XER10 eligibility updates that began on November 19, 2025.

At a glance

Comcast delivers Xfinity TV, Xfinity Internet, Xfinity Home, and Xfinity Voice services through FTTP with an EPON. Comcast has enhanced Hybrid-Fiber Coax (HFC) network by taking the fiber infrastructure to the “last mile." This provides fiber directly into the customer’s home to continue to scale our Fiber offerings. In January 2022, the EPON two-box device solution that replaced the Xfinity Fiber Gateway (XF3) was deployed. In June 2026, EPON with COAM was launched and is now supported.

Notes:

- Comcast introduced the XER10 Gateway for its EPON service areas in August of 2025. As of November 19, 2025, all new EPON customers (except Internet Essentials) will now receive the XER10, regardless of the speed tier they choose (300 Mbps, 500 Mbps, 1 Gig, 2 Gig). Customers requesting voice service will need to still use the Optical Network Unit (ONU) and Technicolor XB7 Gateway setup at

this time. For 2 Gig EPON customers requesting voice service, see the **Xfinity Voice not supported with XER10 Router** section of Introducing the XER10 — The latest Xfinity WiFi Router for EPON internet customers
.

- On accounts with COAM equipment, the customer is solely responsible for the installation, configuration, and ongoing maintenance of their equipment. Technicians are only responsible for company-provided equipment up to and including the ONU. Technicians must ensure a CAT5e or higher Ethernet cable is properly connected to the ONU and left available for customer use. This marks the demarcation point of responsibility. Any assistance provided with customer-owned equipment is strictly at the technician's discretion and is not required. Technicians are not expected to troubleshoot, install, or support customer-owned devices beyond the ONU handoff.

This product offering configuration requires additional steps outside of the traditional XF3 EPON installation guidelines and is a two-box solution:

- An additional component is installed to feed the XB7 or XER10 its service packaging for the customer.
- Next, an Ethernet EPON Optical Node Unit (ONU) is installed and then connected to an XB7 Gateway or XER10 router through Ethernet.

This new solution is the next generation deployment for EPON and will improve the way services are delivered and simplify the EPON deployments in different ways. Some benefits include:

- Customers can optimize the gateway device location without fiber fishing required.
- Builds the foundation to support getting started for EPON customers.

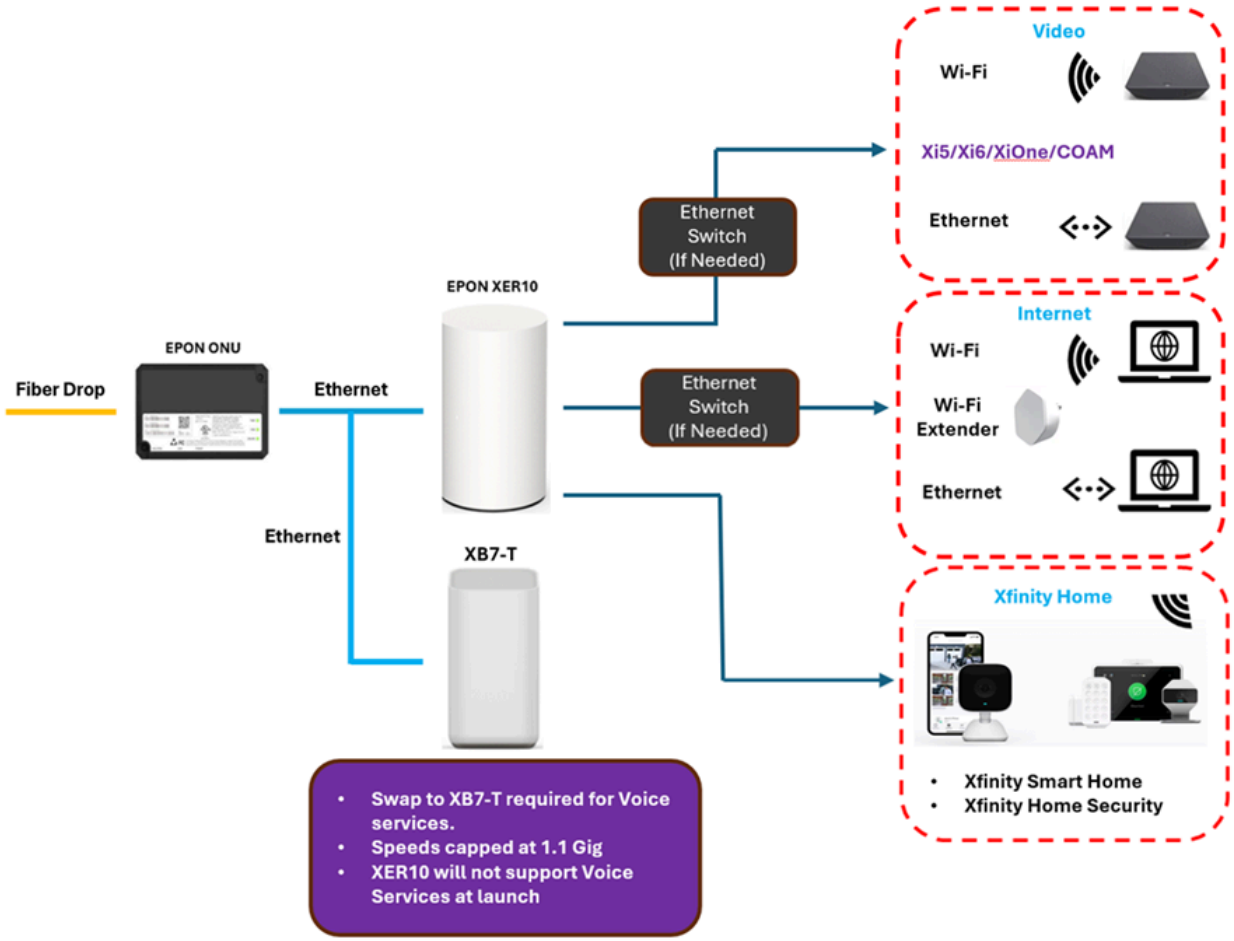
Swapping an XF3 for EPON two-box

The XF3 to ONU upgrade path is available through Tech360, allowing technicians to make an on-the-spot decision to transition a customer from an active XF3 Fiber Gateway to the EPON two-box solution, consisting of an ONU and XER10 router.

Tech360 will guide the technician through the validation and configuration steps required to complete the swap. Once the migration is successful, proceed with the installation and activation of the ONU and Technicolor XB7/Sercomm XER10.

For additional information, see XF3 to two-box on-demand repair swaps job aid.

Residential EPON two-box services delivery diagram



Residential construct for ONU and XB7

Feature	Specifications
Device configuration	• XF-ONU or XF-ONU2 ONU (XF-ONU Nokia Model #:NKFN11AEL, XF-ONU2 Sagemcom Model # SGFN11BEL) • **Note:** Boot file on ONU • Technicolor XB7s (CGM4331COM and CGM433COM) Port 4 is used to connect to the ONU
Device minimum software	• **ONU:** 5.26.46 • **Technicolor XB7:** Minimum software version 7.3
Installation options	• **Non-WiFi Ready:** Pro-install • **WiFi Ready:** Technician pre-install for equipment; customer activation through the Xfinity app
Activation tools Tools that support ONU/XB7 activation	• Tech360 • Einstein 360 • Xfinity app Activation — WiFi Ready Only

Feature	Specifications
Supported products/services	The ONU/XB7 configuration supports all current lines of business and the following products: • Xfinity Internet • IP video and streaming device's video • Voice • Xfinity Home; Pro Protection and Self Protection • Xfinity hotspot • Internet Essentials • WiFi Extenders (formerly known as Pods) • MoCA is supported from the XB7 using the RG6/COAX output up to a maximum of 16 ports (4 x 4 splitter) supporting XiD and other MoCA devices (maximum length is 150 feet)

Feature	Specifications
Unsupported device(s)	• Pro WiFi Extenders (formally known as Storm-Ready WiFi and is only available with XPRO and XB8 or higher, not supported by XB7) • ONUs — Sagecom • Gateway/modems: XB8 — CGM4981COM, CommScope XB7 — TG4482A, XB6, XB3, XB2, and customer-owned modems • Video devices (same as All IP Video): All XGs, RNG-150, CableCARD host devices, and DTAs
Voice Battery Backup	• Optional for residential Xfinity Voice customers • If requested, there needs to be a battery for **both** the ONU and XB7 • See the following articles for more information: • Tech Ops: Xfinity two-box Voice Battery Backup ONU job aid • Tech Ops: Xfinity Voice Battery Backup Case (XBBL) job aid

Residential construct for ONU and XER10

Feature	Specifications
Device configuration	• XF-ONU or XF-ONU2 (XF-ONU Nokia Model #: NKFN11AEL, XF-ONU2 Sagemcom Model #: SGFN11BEL) • **Note:** Boot file on ONU • Sercomm XER10 (SCER11BEL)
Device minimum software	• ONU: 5.26.46 • XER10: min 8.1p8s3 for MACsec
Installation options	• Non-WiFi Ready: Pro install • WiFi Ready: Technician pre-install for equipment; customer activation through the Xfinity app
Activation tools Tools that support ONU/XER10 activation	• Tech360 • Einstein 360 • Xfinity app activation — WiFi Ready only

Feature	Specifications
Supported products and services	• Xfinity Internet • Multi-Gigabit symmetrical speeds need a CAT6 Ethernet cable that cannot exceed 180 feet • IP Video and streaming devices video: Xi6, Xi5, XiOne, and Xumo IP video devices only • Xfinity Home; Pro Protection and Self Protection • Xfinity hotspot • WiFi Extender (formerly known as Pods) • Xfinity approved Zigbee-enabled devices

Feature	Specifications
Unsupported device(s)	• Bridge mode • MoCA • Pro Wifi extenders and Xfinity LTE extenders will not be available at launch • ONUs — Sagecom • **Video devices:** All XGs, RNG-150, XiD, Xi3, CableCARD host devices, and DTAs • Voice services will not be supported at launch ◦ **Option 1:** Stay with ONU/XB7. Speeds will be capped at 1 Gig ◦ **Option 2:** Repackage Xfinity Voice into Xfinity Mobile line and proceed with install

EPON installation details and reminders

EPON installation alerts

	Sagemcom ONU	Nokia ONU
		
BEAD locations	✓	X
Non-BEAD locations	✓	✓

- The Nokia ONU can only be used in non-BEAD locations while the Sagemcom can be used in all locations (see diagram).
- If resetting the ONU, **all** visibility of devices on the premise side of the ONU will be lost, including the Technicolor XB7/Sercomm XER10.
- Technicolor XB7/Sercomm XER10 configurations **must** employ dedicated **(no Ethernet switches or hubs)** CAT5E/6 wire connection between the ONU and the Technicolor XB7/Sercomm XER10.
  - For the Technicolor XB7, use the 2.5G WAN/LAN Port #4 **only** — Identified by an orange marker.
  - For the Sercomm XER10, use the WAN port and CAT 6 Ethernet cable no more than 180 feet.

EPON installation reminders

- **Co-locate the Technicolor XB7/Sercomm XER10 and the ONU** during the activation/provisioning process.
  - Once activated, move the Technicolor XB7/Sercomm XER10 to the location that supports best WiFi coverage for the customer.
  - If there are issues, try to use a known good Ethernet CAT 5E/6 cable during the setup, then switch.

- In the **two-box scenario**, the boot file will be on the **ONU device** instead of the modem.
- Before adding or provisioning in Tech360, make sure to inspect before you connect to ensure that the connector and ONU port are clean, undamaged, and free of debris.
- Ensure light levels are within proper OES standards.
  - **Note:** If the light levels are not within specifications at the tap/nap, follow the RTM process.
- When connecting the fiber to the ONU, make sure you hear a click, and the cable is securely attached.
- Use the ONU serial number to add the ONU in Tech360.
- When swapping the XB7/XER10 or ONU, you should always reboot all devices in sequence. The ONU may hold the old XB7/XER10's IP+ address and rebooting the ONU before connecting the new XB7/XER10 should clear that.
- The ONU must be placed in a higher outlet position than the XB7 or XER10. If the ONU is not in the higher outlet position, it may result in issues with backend tooling and could affect ONU provisioning — leading to poor customer experience. This can only be corrected in CSG or by the technician using Tech360.
- On accounts with COAM equipment, the customer is solely responsible for the installation, configuration, and ongoing maintenance of their equipment. Technicians are only responsible for company-provided equipment up to and including the ONU. Technicians must ensure a CAT5e or higher Ethernet cable is properly connected to the ONU and left available for customer use. This marks the demarcation point of responsibility. Any assistance provided with customer-owned equipment is strictly at the technician’s discretion and is not required. Technicians are not expected to troubleshoot, install, or support customer-owned devices beyond the ONU handoff.

Other EPON installation notes

- Watch Tower now reflects ONU optical levels:
  - **DsRx -17.696** is the ONU **input** light level.
  - **UsTx 6.079** is the ONU **output** light level.

- The Ethernet (minimum CAT5E/6) connection between the ONU and Technicolor XB7/Sercomm XER10 uses the IEEE 802.1AE (MACsec) standards, which specifies a set of protocols to meet the security requirements for protecting data traversing Ethernet LANs.

Inspect before you connect and ONU placement

1. **ONU placement:**
    - Validate a UPC fiber drop connector has been installed all the way to the media smart panel, the fiber house box, or to the centralized point in the home or existing fiber entry point.
        - If a fiber connector is not installed, refer to your local market process or request to install the fiber drop/outlet or the fiber connectors.
    - Outside placement is no longer approved. Indoor placement in a controlled environment is required.
    - All ONUs should be mounted at all locations. Use “Property of Comcast” blue tags on **all** installs.



- **Comcast tag details:**
  - 93475.
  - YO1.
  - Telecrafter Products.
  - 972POCBL.
  - **Tag wording:** *"Property of Comcast. Do not tamper with connections or remove devices."*
  - Blue; 1-inch x 2-inch zip; 3-inch zip style.
  - Order in multiples of 100.
- Technicians must attach a blue **Do Not Tamper** tag to the fiber cable at the ONU/Xfinity Gateway. This prevents customers from unplugging or damaging the connection.
  - **Install the tag:**

    - Attach the tag during installs, service changes, and trouble calls (TCs).
    - Place it at the back of the gateway/router on the fiber optic cable.
- **Apply the tag carefully:**
    - Apply the tag on the fiber connector boot without damaging the cable.
- **Position the tag for visibility:**
    - Ensure the tag faces outward for visibility inside a media distribution panel.
- **Educate the customer:**
    - Explain the importance of not disconnecting the fiber connection.



Tag attached on the boot of the fiber connector



Mounted XF3 w/ tag facing out

2. Once the fiber connector installation has been confirmed, follow the Inspect before you connect FTTx guidelines.
3. If unable to get a pass, and/or light levels do not meet the minimum OES standard:
    - Follow local RTM process to get same-day resolution.
    - Contact a supervisor immediately. You may not be able to proceed.
4. Validate the ONU ports are clean, undamaged, and free of debris.

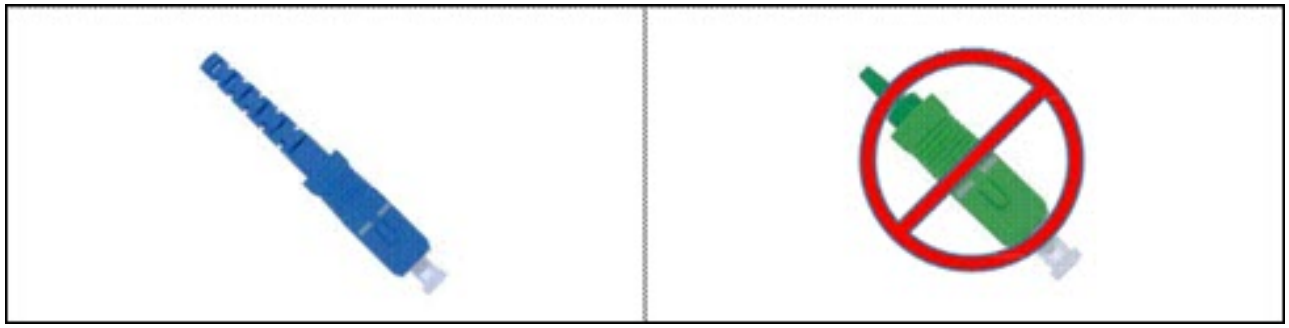
Note: Do not install the green SC/APC fitting into the ONU.

Fiber connections come in two common polishes:

- SC/UPC is **blue** with a flat polish (zero degrees).
- SC/APC is **green** with an angled polish (eight degrees).

The green and angled SC/APC is the most common in our network, but the **blue and flat SC/UPC is what is necessary to connect** to the ONU and OLT with PON.

- The ONU **may not work consistently** with the APC (green/angled) connector.



If the SC/APC is green connector is installed on the fiber drop, there are two methods of addressing this issue:

1. Install a hybrid jumper and coupler to change the interface from SC/APC to SC/UPC, or
2. Cut off the SC/APC and install an SC/UPC connector using one of the approved mechanical or splice-on connectors mentioned in **section 5.4 – Interior Fiber Drop** in the **EPON two-box deployment installation update**.
   - SC/APC to SC/UPC jumper and coupler:



   - SC/APC to SC/UPC jumper and coupler

Manufacturer name	Manufacturing part number	Item description	CIFA number
CommScope	BC1000-AS	Bulkhead/SC/APC connector with clip	38442
CORNING CABLE SYSTEMS INC.	MAR1SURABMN0.5OMYL	Fiber Jumper, SM, Simplex BI yellow riser, 1.2mm, SC/UPC-SC/APC with short boots, 0.5 meter	172328

Guidelines for installing the XF — ONU in a media panel

- **Media panel purpose:**
  - Media panels organize and manage wiring in homes and MDUs.
- **Compatible boxes:**
  - Plastic and metal media boxes can be used.
- **ONU placement:**
  - Position the ONU for easy access to the fiber fitting and power port.
- **Cable protection:**
  - Follow the minimum bend radius for fiber and power cables.
- **Orientation flexibility:**
  - The ONU can be installed in any direction to save space.

- **Power source:**
  - Use the included adapter and plug it into the panel's AC outlet.
- **Fiber transition point:**
  - Add a transition point to reduce wear on the incoming fiber and connector.


DO NOT REMOVE
FROM UNIT



7-2 Installation of fiber transition outlet

Identify a job as an ONU and Technicolor XB7 or Sercomm XER10 install in Tech360

1. Go to Tech360.
2. Review the job under **Devices to Install**.
    - The verbiage will indicate if an XB7 or an XER10 is to be used.
3. If not, check the **Package Details**.
    - It will be indicated by what speed tier the customer is getting.

As mentioned prior, all new EPON customers (except Internet Essentials) will now receive the XER10, regardless of the speed tier they choose (300 Mbps, 500 Mbps, 1 Gig, 2 Gig).

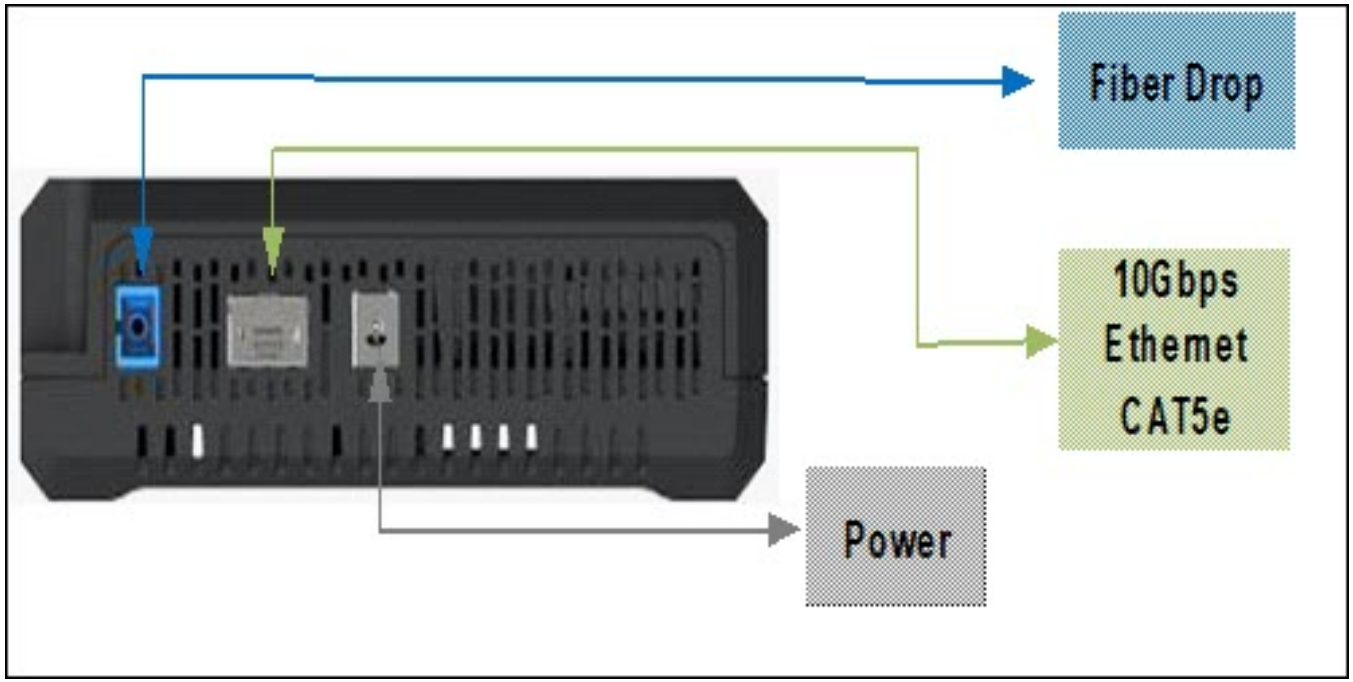
- **ONU already on-site**

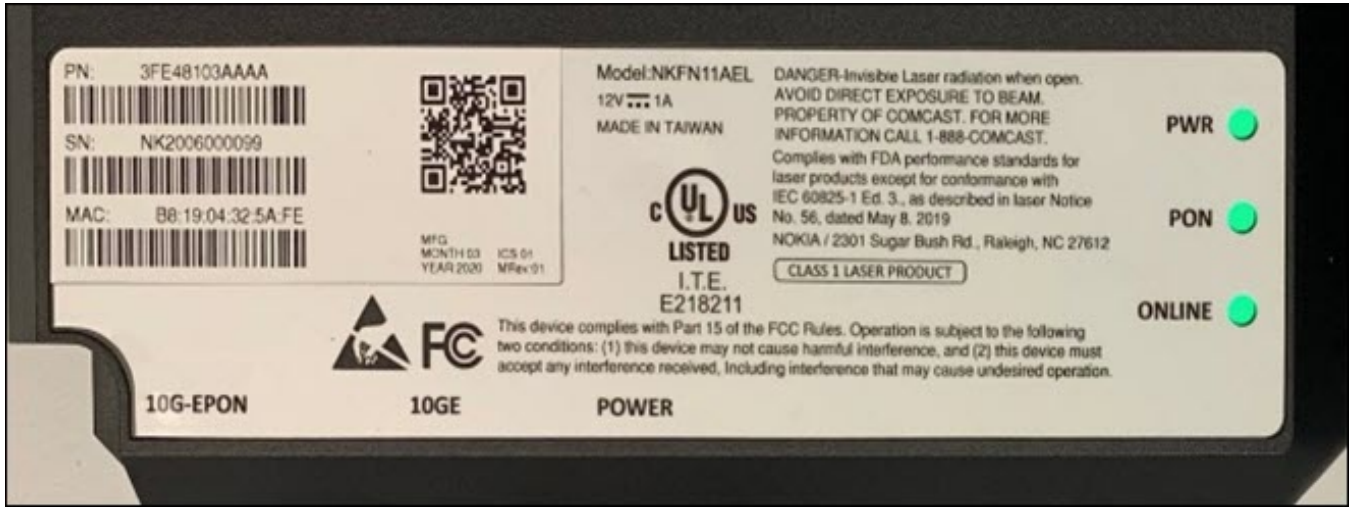


- **Net new install**

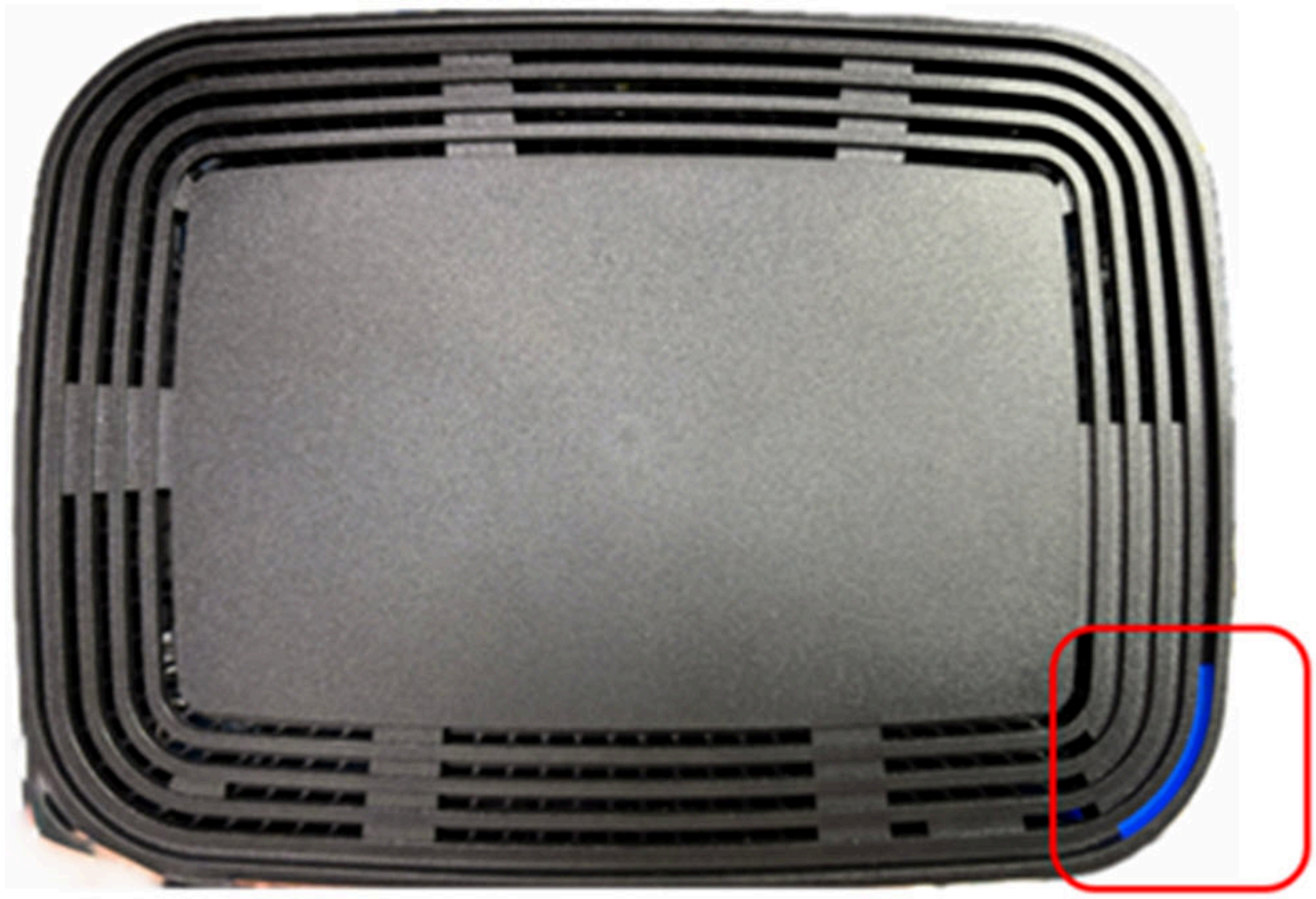


EPON-ONU (XF-ONU Nokia NKFN11AEL or XF-ONU2 Sagemcom SGFN11BEL) installation



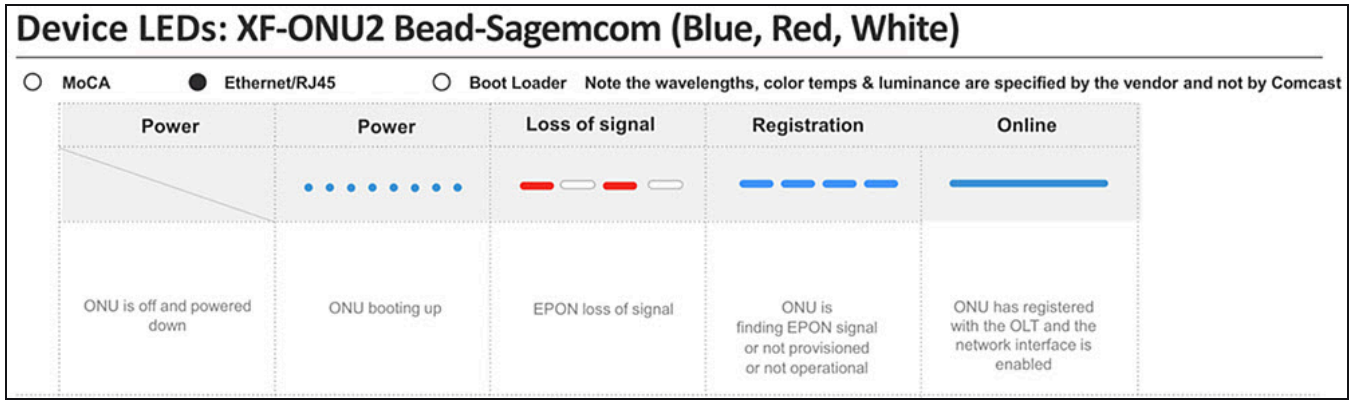


XF-ONU (Nokia NKFN11AEL)





XF-ONU2 (Sagemcom SGFN11BEL)



ONU and Technicolor XB7 installations

XB7 preliminaries for EPON

Important: With the release of RDK version 7.3p5s1 on Technicolor XB7s, manually setting the XB7 WAN Network to **Ethernet** is no longer required. Some devices already within the supply chain flow to the field, on pick walls, totes, trucks, etc. may not have the updated version of code. If the XB7 does not become operational within five minutes, it may still be on the older firmware and will still require the manual setting of the WAN Network to **Ethernet**.

If you have issues with your installation, for example the XB7 continues to blink amber, the ONU has all green LEDs and is connected directly to the XB7 to port 4, you may still need to proceed with manually setting the eWAN mode to **Ethernet**. For instructions, review how to manually set the XB7 or XB8 into eWAN mode.

Currently, the Technicolor XB7 is only supported. The CommScope XB7 is **not** supported.

There are two ways to identify a Technicolor XB7:

- By the tag on the bottom of the device.
  - **Technicolor XB7 PN:** CGM4331COM
  - **CommScope XB7 PN:** TG02DCW4482CT

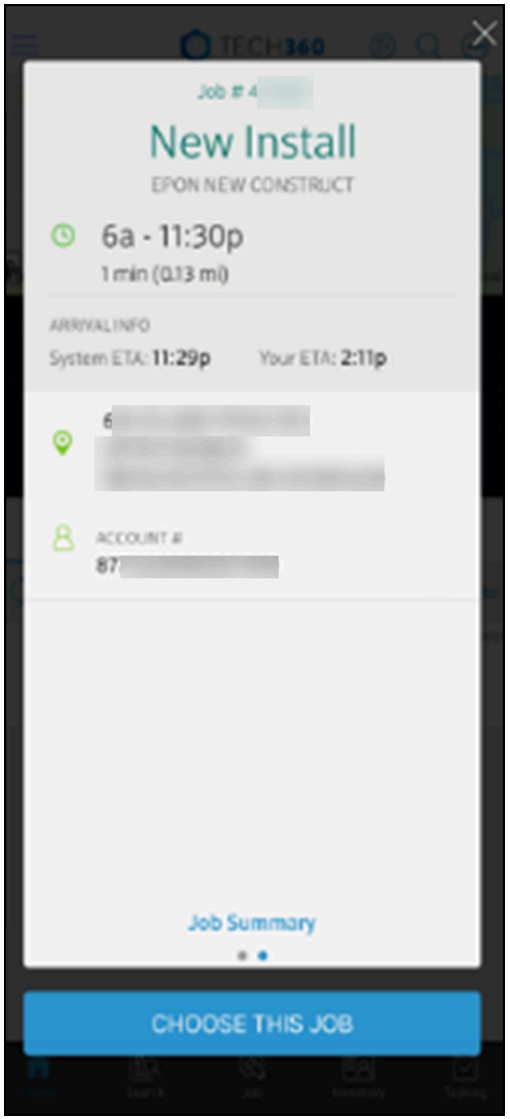
- On the back of the XB7, just below the WPS button is the identifier.
  - **XB7-T for Technicolor**
  - **XB7-CM for CommScope**

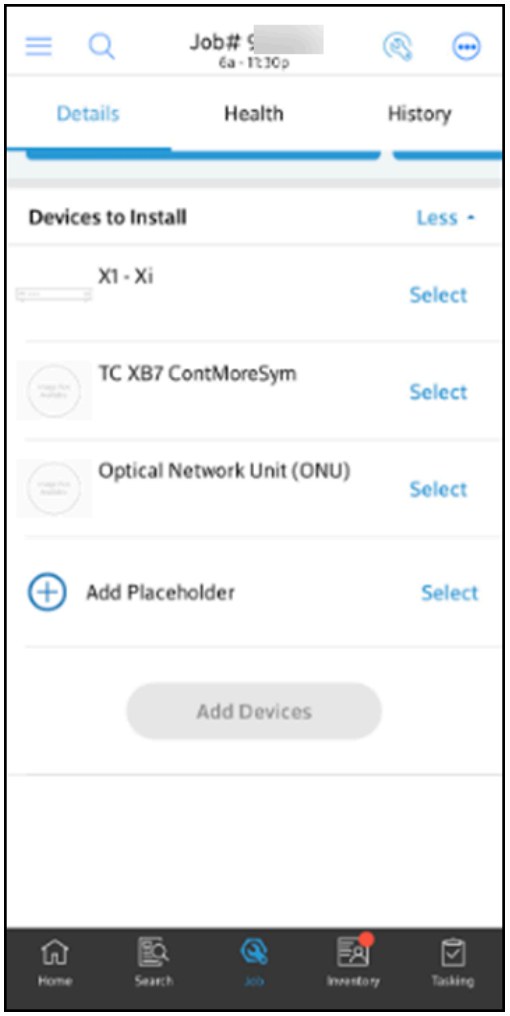


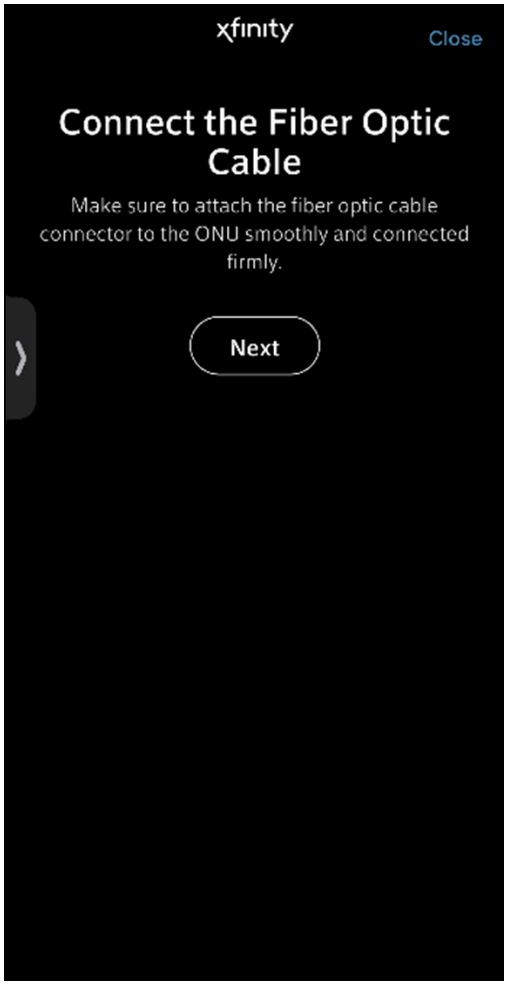


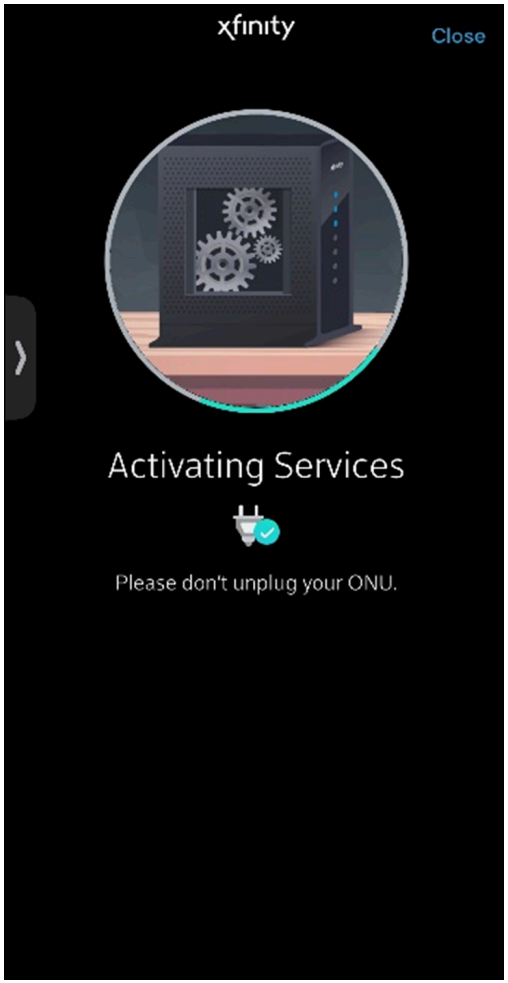
New ONU (Nokia NKFN11AEL) install

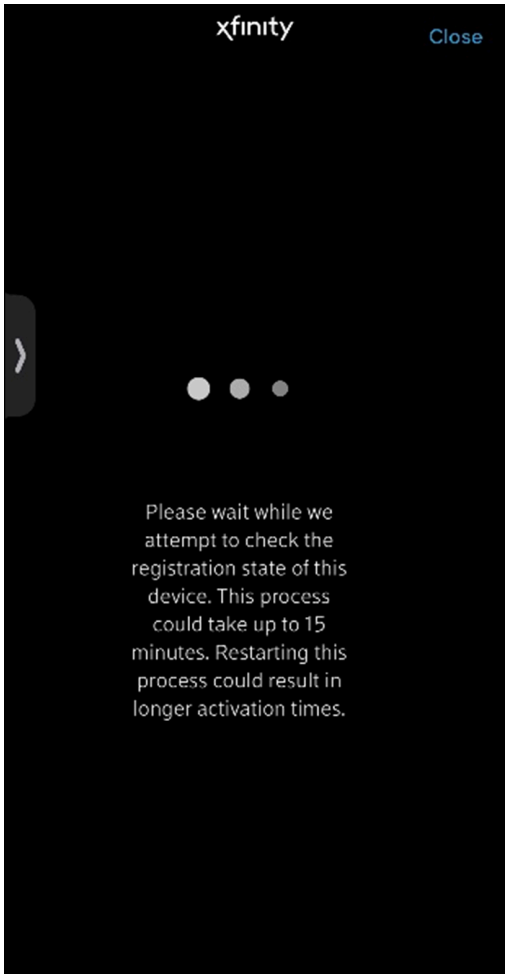
1. After powering up the ONU, wait for the ONU to complete its registration process until all three LEDs are **solid green**.
2. Add your devices in Tech360 to the customer's account.
3. Initiate and activate services in Tech360 following the Tech360 prompts.
   - Tech360 will guide you through provisioning the ONU unit onto the customer's account.


TECH360
Job # 4
New Install
EPON NEW CONSTRUCT
6a - 11:30p
1 min (0.13 mi)
ARRIVAL INFO
System ETA: 11:29p
Your ETA: 2:11p
ACCOUNT #
Job Summary
CHOOSE THIS JOB


Job#
Details
Health
History
Devices to Install
Less
X1 - Xi
Select
TC XB7 ContMoreSym
Select
Optical Network Unit (ONU)
Select
Add Placeholder
Select
Add Devices
Home
Search
Inventory
Tasking


xfinity
Close
Connect the Fiber Optic Cable
Make sure to attach the fiber optic cable connector to the ONU smoothly and connected firmly.
Next


xfinity
Close
Activating Services
Please don't unplug your ONU.



ONU (Nokia NKFN11AEL) already on-site

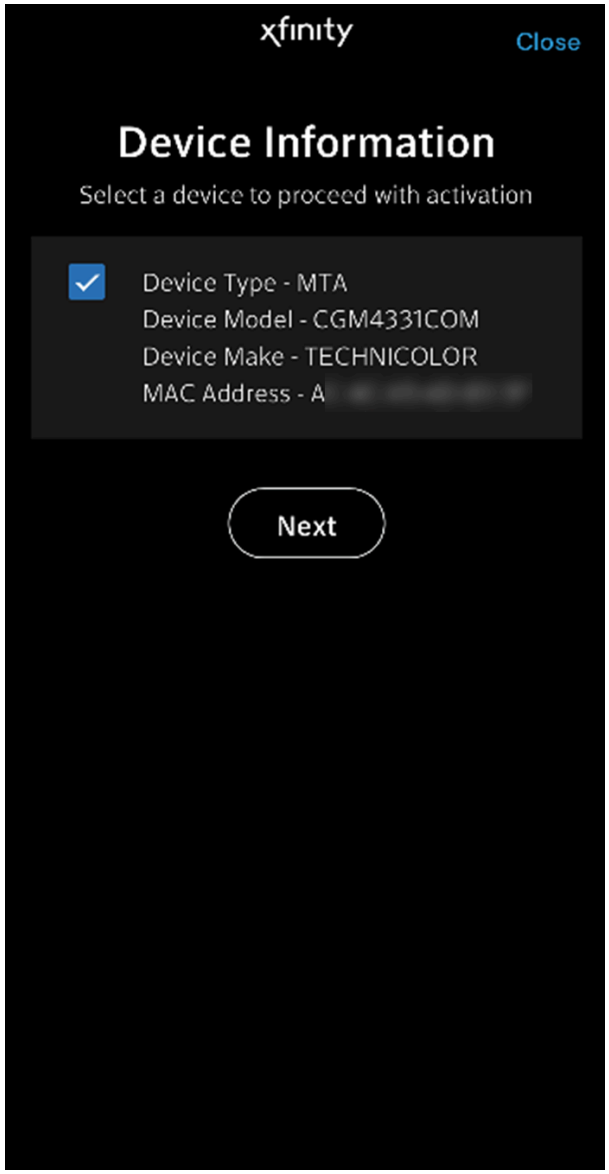
When an account is disconnected, the ONU should always remain on-site for the next customer to connect at that location.

1. You should see the ONU placeholder pre-populated with the ONU serial number in Tech360.
2. Add a new Technicolor XB7.
3. Select **Add Devices**.
    - Both devices should move to the **Activated Devices** section.

- Tech360 will guide you through the install and activation like a regular new install.

If the ONU is on-site and already active, you should not expect to see it in Tech360 and would only need to activate the Technicolor XB7.





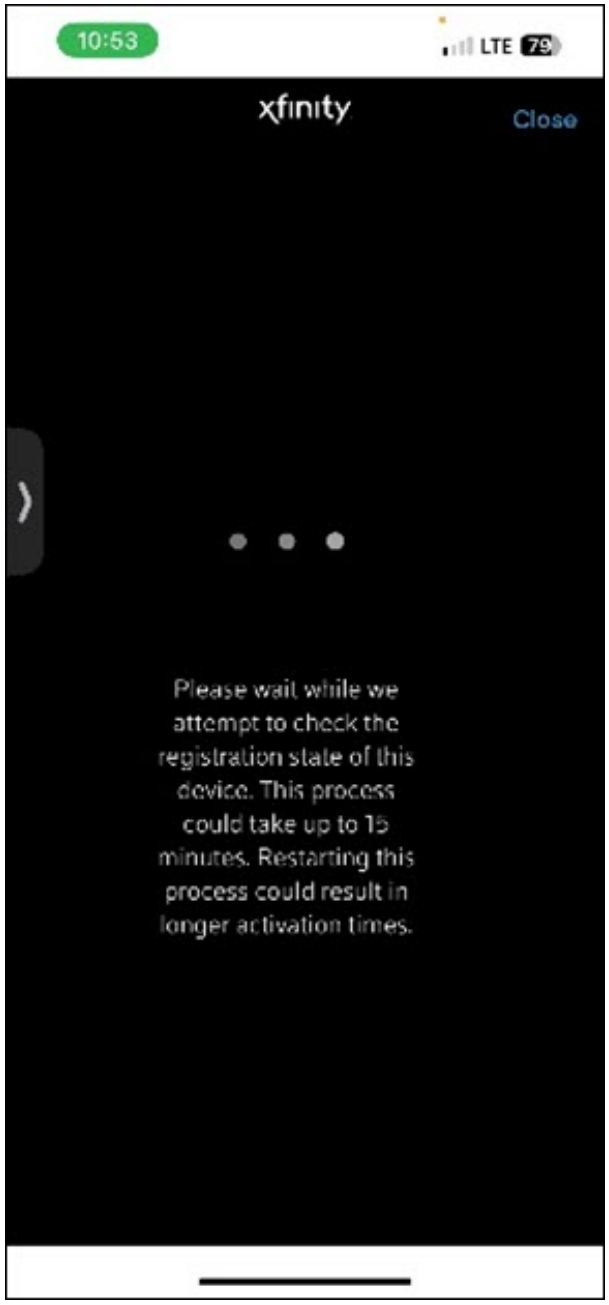
Technicolor XB7 Gateway installation and activation for EPON

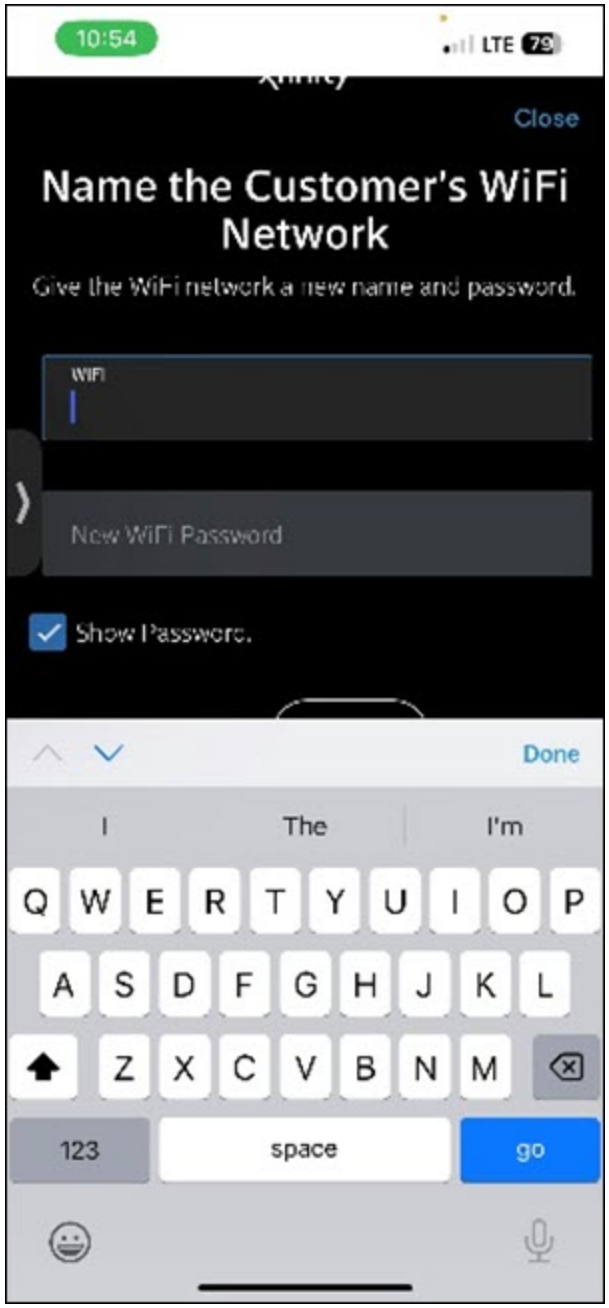
1. When instructed to do so, connect the Technicolor XB7 and ONU with Ethernet CAT5E/6 cable to port 4, positioned above the power plug with an **orange** marker.

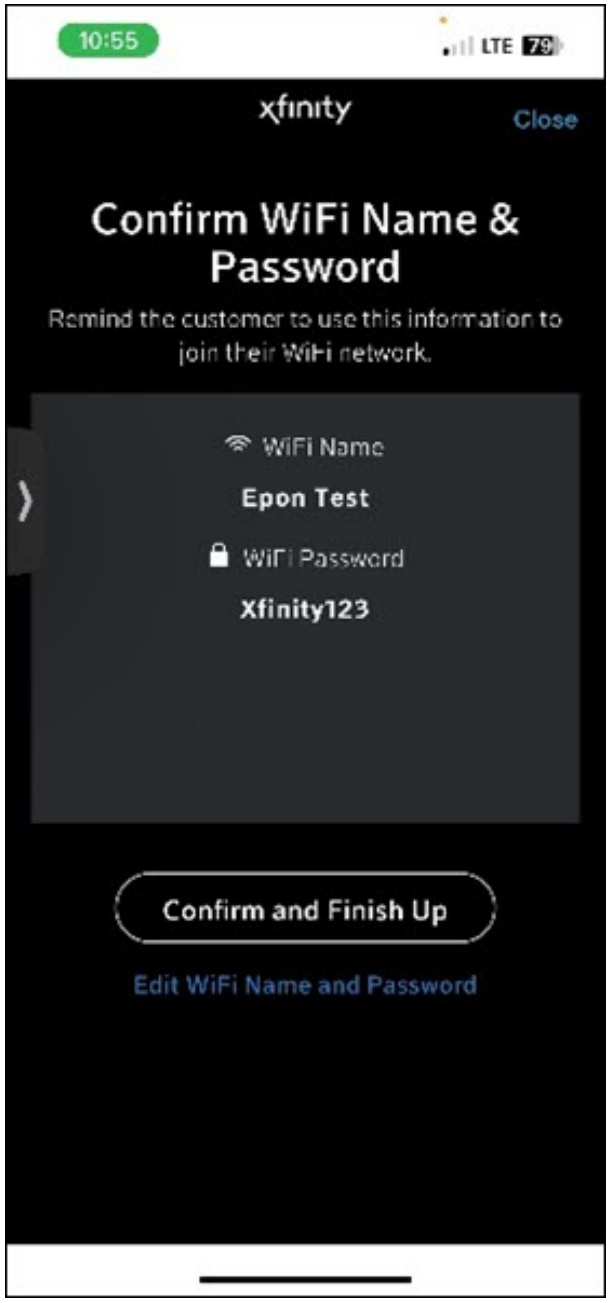
2. Begin the Tech360 workflow to initiate and start up sequences, allowing the gateway to activate.
    - **Note:** Be patient. Once connected, the Technicolor XB7 may take up to seven minutes (or longer) to come online.
        - Do **not** reset or power cycle the gateway until this process is completed.
        - The XB7 will blink amber until it transitions into WAN mode and gets an IP address from the ONU.
            - If this takes more than 10 minutes, reboot the ONU and verify that the eWAN mode is set to Ethernet.
3. Before moving on to setting up the WiFi name and password, wait until the Technicolor XB7 transitions from blinking amber to blinking white.

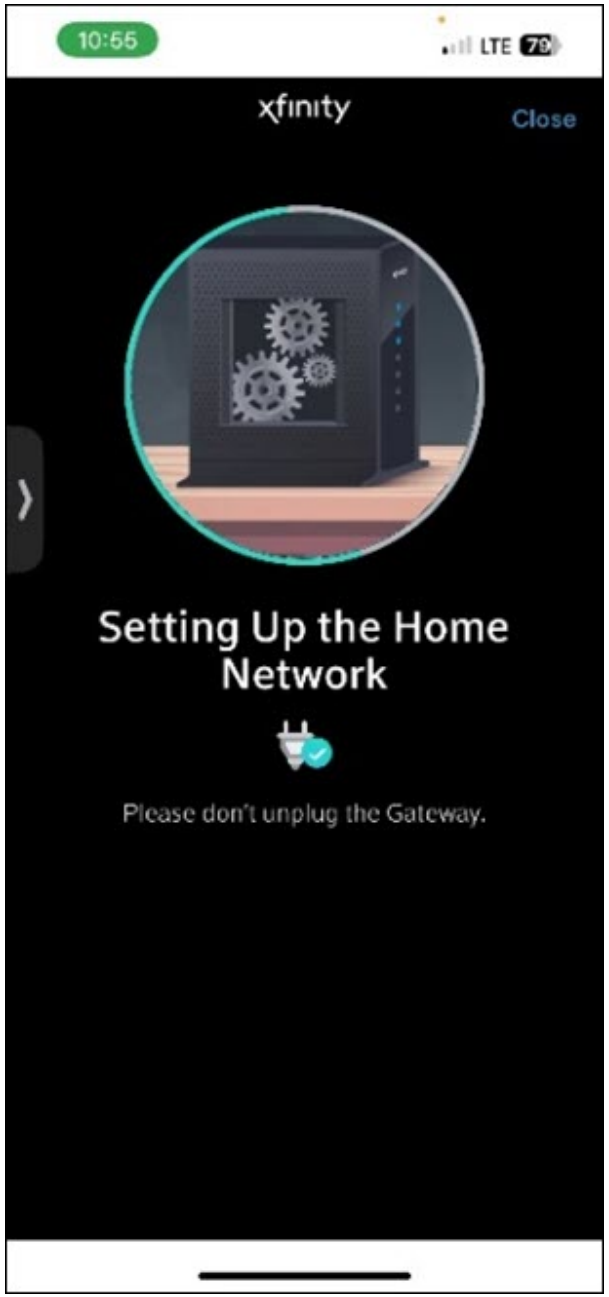
When these steps are complete, the Technicolor XB7 LEDs should be solid white.


10:53
LTE 79
xfinity
Close
Connect the Ethernet Cable
Connect the ethernet cable between the Gateway and the ONU. Make sure the ethernet cable is plugged in to the correct port of the ONU and Gateway.
Next

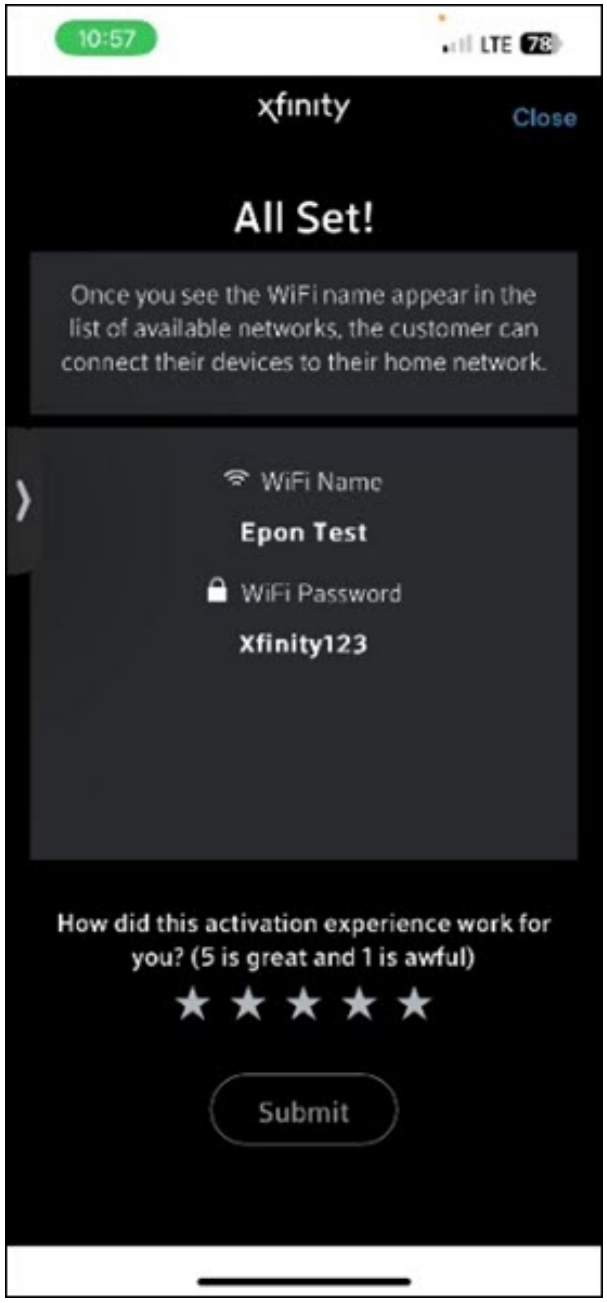

10:53
LTE 79
xfinity
Close
Please wait while we attempt to check the registration state of this device. This process could take up to 15 minutes. Restarting this process could result in longer activation times.


10:54
LTE 79
Close
Name the Customer's WiFi Network
Give the WiFi network a new name and password.
WiFi
New WiFi Password
Show Password.
Done
I
The
I'm
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
go


10:55
LTE 79
xfinity
Close
Confirm WiFi Name & Password
Remind the customer to use this information to join their WiFi network.
WiFi Name
Epon Test
WiFi Password
Xfinity123
Confirm and Finish Up
Edit WiFi Name and Password


10:55
LTE
xfinity
Close
Setting Up the Home Network
Please don't unplug the Gateway.


10:56
LTE 78
Setting Up the WiFi...
This can take up to 10 minutes.
Close
Almost Activated
Once you see the WiFi name appear in the list of available networks, the customer can connect their devices to their home network.
WiFi Name
Epon Test
WiFi Password
Xfinity123
Please don't unplug the Gateway.


10:57
LTE 78
xfinity
Close
All Set!
Once you see the WiFi name appear in the list of available networks, the customer can connect their devices to their home network.
WiFi Name
Epon Test
WiFi Password
Xfinity123
How did this activation experience work for you? (5 is great and 1 is awful)
Submit

ONU and Sercomm XER10 installations

XER10 preliminaries

The XER10 provides significantly better performance and a smoother installation process for Xfinity EPON residential customers.

See the XER10 features and interface capabilities below.

Feature and interface capability	Specifications
SoC	BRCM 4916
WiFi	WiFi 7 Tri-band configuration (4 x 4): 2.4 GHz - BRCM6726; 5 GHz - BRCM 6726; 6 GHz - BRCM 67263
IoT	Zigbee, Bluetooth
Flash	4 GB eMMC
DRAM	2 GB DDR4
Ethernet	One 10 Gbps WAN port One 10 Gbps LAN port Three 1 Gbps LAN ports
Voice	One RJ-11 port

XER10 – Xfinity 10G Ethernet Router

Features

SoC	BRCM 4916
Wi-Fi	Wi-Fi 7 Tri-band configuration (4x4) 2.4GHz- BRCM 6726 5GHz- BRCM 6726 6GHz- BRCM 67263
IoT	Zigbee Bluetooth
Flash	4 GB eMMC
DRAM	2 GB DDR4

Interfaces

Ethernet	[1] 10Gbps WAN port [1] 10Gbps LAN port [3] 1Gbps LAN ports
Voice	[1] RJ-11 port



Before starting the installation:

1. Inspect the onsite Ethernet cable to verify CAT5e or CAT6.
2. Once the cable category is determined, find the best location for the XER10 and discuss it with the customer.
    - **Option 1:** The media panel should be the primary installation location. If door extender is required, ensure it is installed and door closes.
    - **Option 2:** Use in home CAT wiring. Tone and terminate the ethernet, and locate the device in the unit using in home CAT wiring. For **2 Gig services, CAT6 no more than 180 feet** must be used.
        - If option 1 and 2 are not available, escalate to your supervisor.

No XER10 or ONU onsite — New customer

- It is important to follow the device installation instructions outlined in the two-box job aid, including the XER10 router factory reset process while not connected to a DOCSIS

network (no coaxial connection required).

- **To factory reset the XER10**, hold the reset button on the back until the LED on the front flashes green, then release and allow the device to boot up. This process should take 15-30 seconds to complete.
    - Install the ONU.

- Connect the XER10 to the ONU using the WAN port (bottom port of the device).


Reset
Phone
1 Gb
1 Gb
1 Gb



- Follow the step-by-step instructions in Tech360 to install and activate both devices.

ONU onsite but no XER10 — New customer, existing customer

- Connect the XER10 to the ONU using the WAN port (bottom port of the device).
- Follow the step-by-step instructions in Tech360 to complete the installation and activation process.

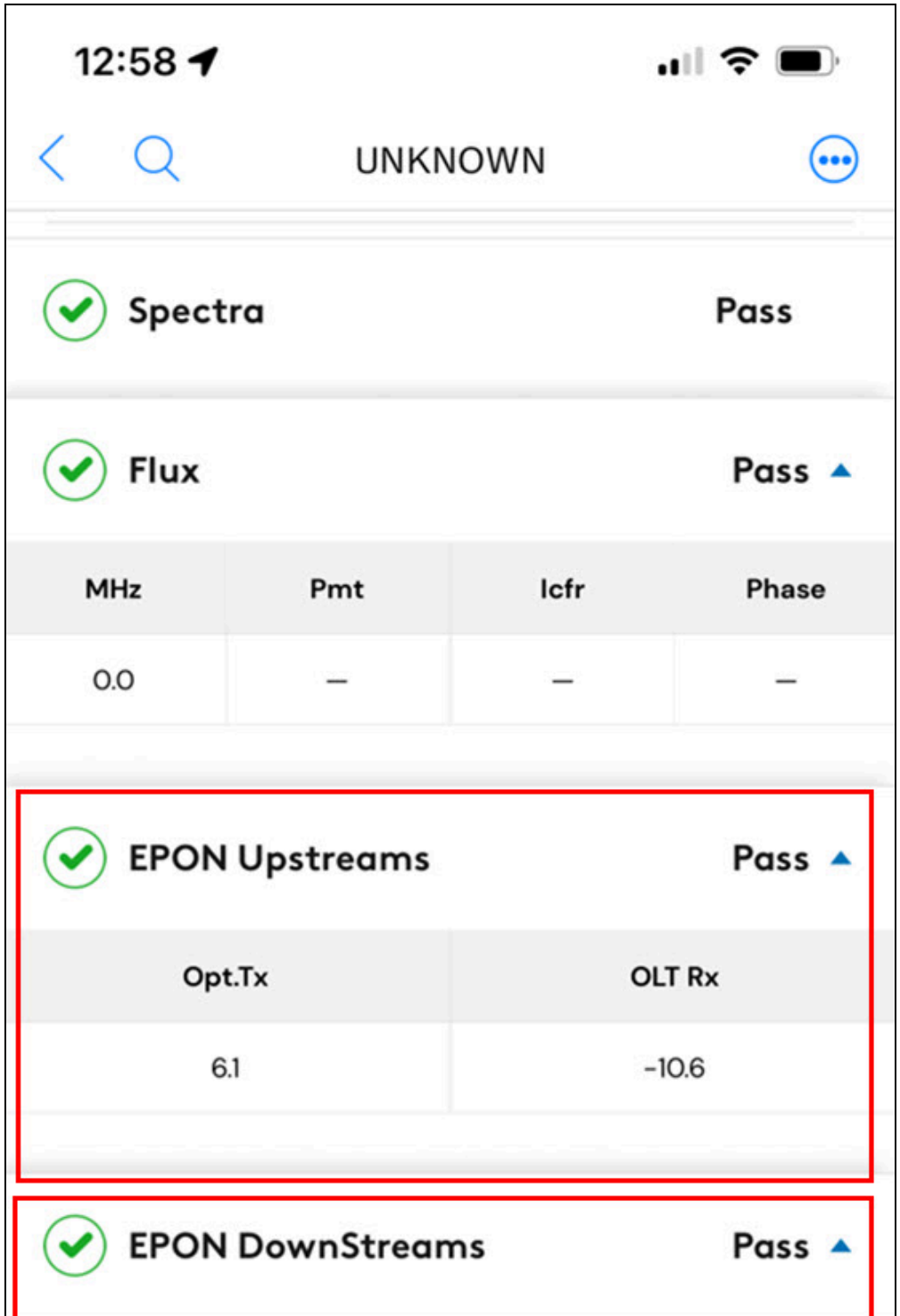
Complete installation of other customer services and equipment

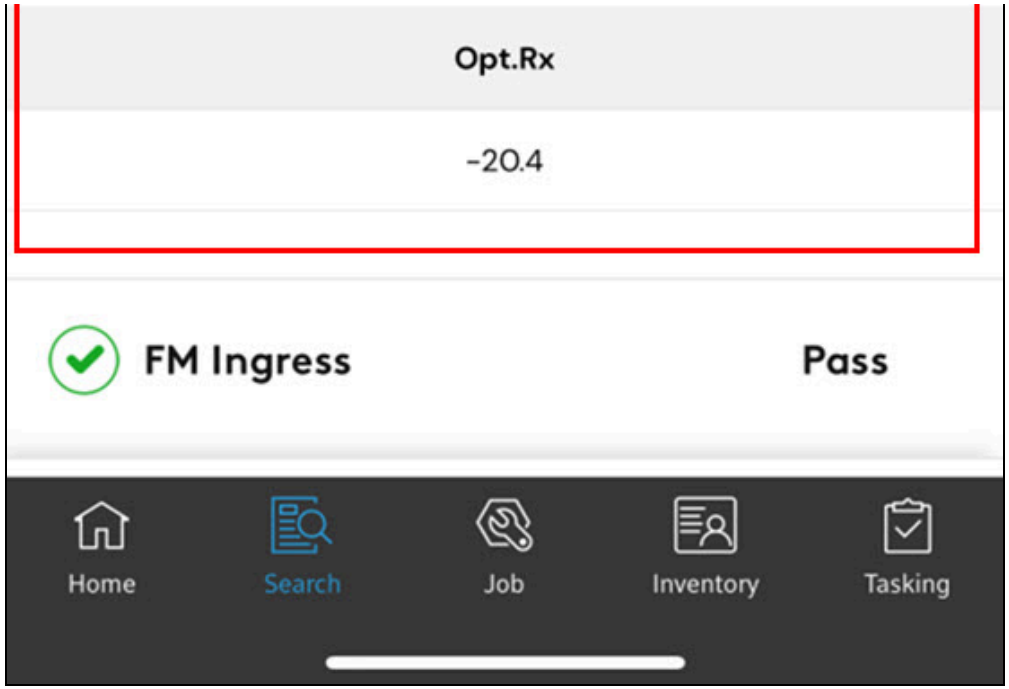
If the ONU is reset during activation, the XER10 LED will go to a **blinking amber** — this is expected.

Once the Technicolor XB7/Sercomm XER10 is activated and installed, follow business as usual installation guidelines specific to the set-top box combination (MoCA or WiFi), Xfinity Home, and Xfinity Voice that you will be installing on the customer's account.

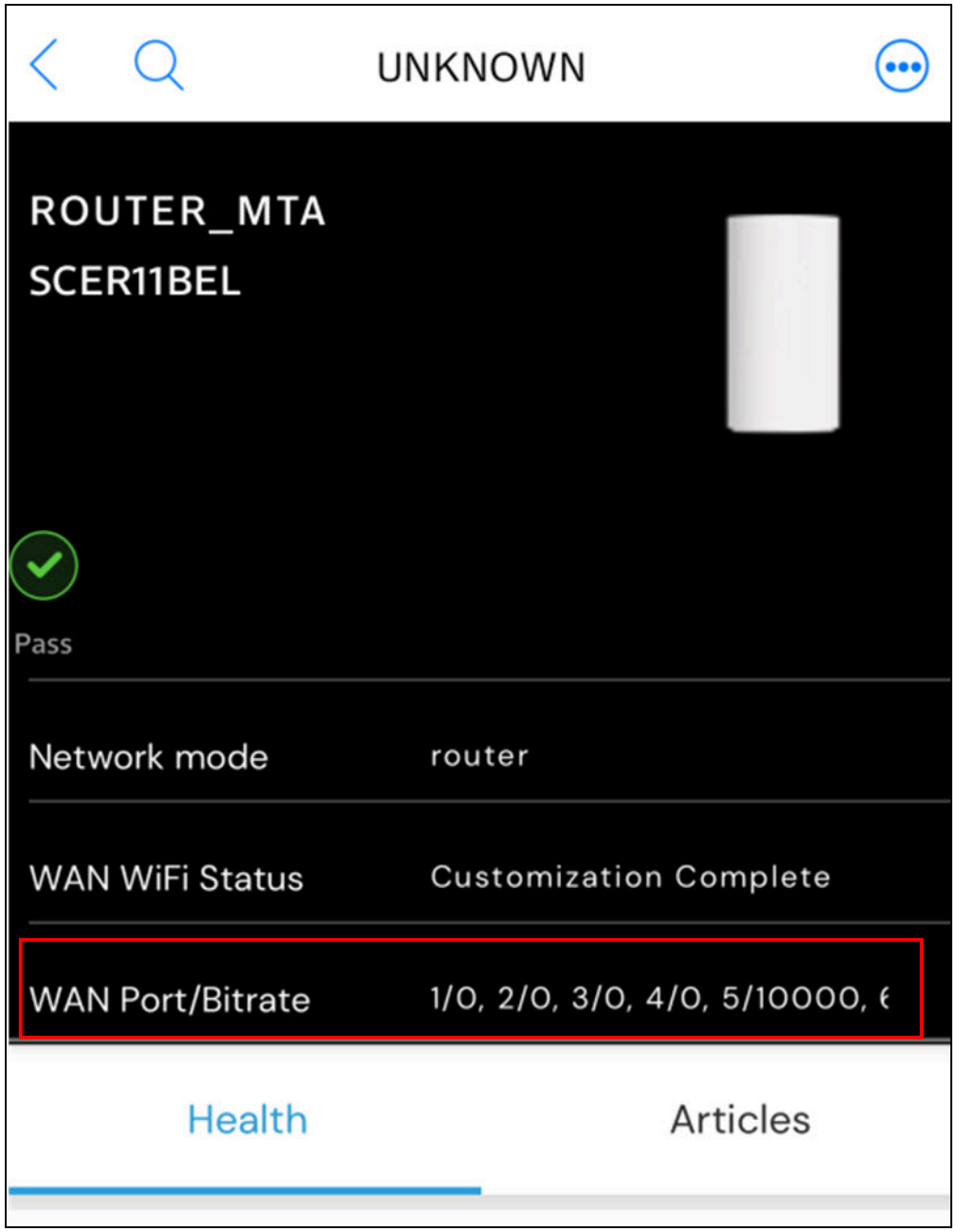
To verify if activation was successful on Tech360, technicians should check device health of the ONU after running the Premise Health Test (PHT). In device health, there should be green check marks for multiple categories including **Flux**, **EPON Upstreams**, and **EPON**

Downstreams. If these are failing with a **red X**, follow the same guidelines found above in Inspect before you connect and ONU placement.


12:58
UNKNOWN
Spectra
Pass
Flux
Pass
MHz
Pmt
Icfr
Phase
0.0
—
—
—
EPON Upstreams
Pass
Opt.Tx
OLT Rx
6.1
-10.6
EPON DownStreams
Pass

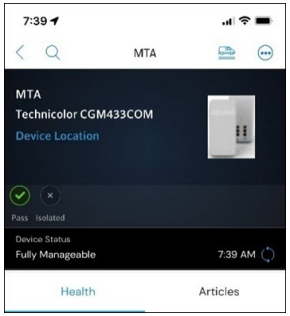


Device health for the XB7/XER10 will show a MAC address, as well as a **Pass** status under WAN. Also, **WAN Port/Bitrate** is visible, allowing for a technician to certify the correct download speeds from the ONU to the XB7/XER10 and what port it is active.


UNKNOWN
ROUTER_MTA
SCER11BEL
Pass
Network mode
router
WAN WiFi Status
Customization Complete
WAN Port/Bitrate
1/0, 2/0, 3/0, 4/0, 5/10000, €
Health
Articles

Tech360 error messages for EPON

Use case	Error message	Troubleshooting steps
Tech360 displays the device is not fully manageable	***Device is not Fully Manageable*** *Please give us a call to Trouble Support Center (TSC) so we can take care of this as quickly as possible.*	Although a modem may display the blinking white LED sequence, indicating it is ready to be activated, the device must be fully registered as online with the Comcast backend systems. • Before activating, confirm the device status in Tech360 is **Fully Manageable**. • If you attempt to activate a modem before it is Fully Manageable, you may get the **Device is not Fully Manageable** error. • This error occurs 100% of the time if the Technicolor XB7 is running 4.11 firmware. • Once the firmware reaches 5.6, you

Use case	Error message	Troubleshooting steps
		can check the XB7 status by clicking on the modem and checking the status.  *Click image to enlarge.* • Before calling the TSC or Testing Center of Excellence (TCOE), wait five minutes, then retry. • If unsuccessful, reboot and retry. • If still unsuccessful, **swap** the device with another device if possible.

Use case	Error message	Troubleshooting steps
Tech360 error when adding ONU	***Alert!*** *Error updating the following equipment…*	• Confirm entry of serial number and MAC addresses. • Call TSC/TCoE.
Tech attempted to install an incompatible device	***Hold Up*** *This device is not compatible with ONU. Please use a Technicolor XB7.*	Install a Technicolor XB7.
Tech360 “Not in inventory” error	***Error Occurred*** *The equipment you are trying to add is not in inventory. Please contact CommOps for assistance (ACP_EQP_Serial: _ _ _ _ equipment not in inventory).*	• Ensure the customer premise equipment (CPE) has been properly received in your inventory. • If not, you need to use a new device.

Use case	Error message	Troubleshooting steps
Outlet reorder error	***Error Occurred*** *There was a problem reordering the devices. Please try again. [WOSE0033.2076]*	If/when you move the ONU around, and the system tries to assign it an outlet designation, it fails and generates the error shown. If you receive the error: • Acknowledge the error, then cancel out of the reorder screen. To reorder the outlets, contact TSC/TCoE.

ONU device troubleshooting

ONU offline

If the ONU won't come online or all three LEDs are not green:

1. Inspect the fiber feed.
   - Clean the fiber feed as needed.
   - Follow the inspect before you connect and ONU placement guidelines during inspection
2. If the ONU still won’t connect, follow business as usual steps to swap it for a new ONU.

ONU device access and reset

- ONU won't appear in (CSG) or (ASCR).
- Customers can't reset ONU remotely. Technician visits are required for issues.
- ONU is free and stays on site.

Two-box setup and boot file

- The boot file is stored on ONU, not on the Xfinity Gateway.
- View the **boot file** in Einstein 360 under **ONU device**.

Reboot times and reminders

- The XB7 reboot may take up to five minutes due to DOCSIS/EWAN search.
- ONU reboot takes less than 60 seconds.
- Tech360 prompts technicians to wait for reboot completion.

ONU inventory and swap

- ONU must be in **Equipment Inventory System (EIS)** before installation.
- ONU swap shows old and new ONUs in Tech360 for 30–40 seconds; old then disappears.

Troubleshoot EPON installation issues

Troubleshoot ONU and XB7 installation issues

XB7 offline/amber LED

If the XB7 has an amber LED:

1. Power cycle the device.
   - Disconnect the power for 10–20 seconds.
   - Plug it back in.
2. Next, check the **WAN setting**. Ensure it’s set to Ethernet.

3. If you still can't get a blinking white LED, follow business as usual steps to swap for a new XB7.

Customer Defined Voice (CDV) activation issues

If no dial tone:

1. Power cycle the XB7.
2. The reboot takes five or more minutes.
3. Wait 5–10 minutes post reboot and retry.
4. If unresolved, repeat or contact TSC/TCOE for an XB7 swap.

Troubleshoot ONU and XER10 installation issues

Flashing amber

If the LED is still flashing amber after three to five minutes, the XER10 is still trying to register and establish a downstream link.

Potential causes:

- ONU LEDs are not all locked on and green.
- Incorrect firmware loaded on the XER10 (min 8.1p8s3 for MACsec).
- Ethernet cable is not connected to the WAN port on the XER10.
- Bad ethernet cable or connections.
- Loose Ethernet cable connection at the ONU or XER10.
- Issue with ONU or XER10 device.

Troubleshooting steps:

1. Verify the ONU LEDs are all **green** (if not see ONU device troubleshooting).
    - Verify the incoming light levels to the ONU.
2. Make sure the Ethernet cable is securely and directly connected to the RJ45 WAN Port of the XER10, as well as the 10 Gig Ethernet port of the ONU. **Change ethernet patch cable or connections** if needed.

3. Power cycle the ONU and XER10.
    - Power down both the ONU and XER10.
    - First, power up the ONU and allow all LEDs to go solid green.
    - Then, power up the XER10 and wait for the LED to either flash white or go solid white. Then, continue with the installation.
        - See ONU LEDs states.
4. If all the above troubleshooting steps fail, contact TSC/TCOE or follow your current market process as directed by your leadership to swap out the XER10 and ONU and re-install.

Solid red or blinking amber, then solid red

The LED may stay solid red, or the LED may go blinking amber then solid red.

This is an error and can mean no internet connectivity (HW, SW, WiFi, Ethernet failure, or any impairment in loss of IP connectivity other than loss of block sync), AC, and battery power.

Troubleshooting steps:

1. Verify the ONU LEDs are solid green.
    - See ONU device troubleshooting.
2. Check there are no outages in the area you are working in.
3. Check fiber light levels coming into the premises and at the fiber tap.
4. Test and validate the fiber connector (blue SC/UPC) isn't damaged or dirty.
5. Verify that the fiber connector is connected and seated properly on the ONU (You should flee it click when connecting).
6. Test and validate the Ethernet cable and connections at the WAN port of the XER10 and the 10 Gig port of the ONU.
    - These should also click when seated properly in the ports.
7. Reboot the XER10.

To factory reset the XER10:

1. Hold the **reset** button on the back of the XER10 until the LED on the front flashes green.
2. Then, release.
3. If it does not recover on reboot, contact TSC/TCOE or follow your current market process as directed by your leadership to swap the XER10.

What do the ONU LED lights mean

Power LED	PON LED	Online LED	ONU state/troubleshooting steps
Green	Green	Green	ONU is powered on and ready to be connected to the XB7/XER10 (if not already). • The LED will be on even if you are not connected to the XB7/XER10. • All LEDs can be green, regardless of what type of boot file it has (real service or walled garden).

Power LED	PON LED	Online LED	ONU state/troubleshooting steps
Green	Green	Blinking green	Code download is underway. No action is needed. When complete, proceed with XB7/XER10 installation. When properly connected to the fiber, the ONU will download the correct boot file and firmware. The ONU may reboot after the download.

Power LED	PON LED	Online LED	ONU state/troubleshooting steps
Green	Off	Off	**Connectivity issue:** • ONU is not detecting the proper light from the OLT. • Try rebooting the ONU. • Verify if there are any outages in the area. • Check all fiber connections, clean if needed. • Perform business as usual checks on both ends of the fiber coming into the premise including the connector. • Verify the light levels if the light levels are within specification. • If there are no outages and the fiber connections are clean and good, and the ONU reset did not resolve the issue, contact TSC/TCOE or follow your current market process as directed by your

Power LED	PON LED	Online LED	ONU state/troubleshooting steps
			leadership to remove and replace the ONU then restart the installation.

Power LED	PON LED	Online LED	ONU state/troubleshooting steps
Green	Green	Off or blinking green	**Possible issues:** • IP connectivity to the OLT is interrupted. • ONU may be misprovisioned. • Boot file is corrupt and not supported at this OLT. • Try rebooting the ONU. • Verify if there are any outages in the area. • Check all fiber connections, clean if needed. • Perform business as usual checks on both ends of the fiber coming into the premise including the connector. • Verify the light levels if the light levels are within specification. • If there are no outages and the fiber connections are clean and good, and the ONU reset did not resolve

Power LED	PON LED	Online LED	ONU state/troubleshooting steps
			the issue, contact Divisional TSC/TCOE or follow your current market process as directed by your leadership, using Einstein 360, they should: • Verify they can see the ONU. • Verify the boot file to make sure it matches the service tier. • Reset the ONU. • Re-provision the ONU. • Swap the ONU.

Technicolor XB7 and Sercomm XER10 LED meanings

For device lights meaning, see What the lights on your Xfinity Gateway mean.

Reporting a Tech360 issue

If you encounter an issue while using Tech360, use the **Report a Tech360 Issue** feature. This feature gets the issue in front of the Tech360 team and will allow them to pull timely

logs as needed.

Switching from Radio Frequency over Glass (RFoG) to EPON two-box

- Switching from RFoG to EPON two-box will require technicians to replace the customer's existing XB Gateway and QAM TV Boxes with an ONU, Technicolor XB7 Gateway/Sercomm XER10 router, and IP video TV Boxes (Xi5/Xi6).
  - **Note:** XiDs are still compatible with the XB7.
- If QAM TV Boxes are present, they must be swapped for IP Video TV Boxes.
  - **Note:** If the customer already has Xi5 or Xi6 devices, no swap is necessary.

All QAM TV Boxes must be swapped in Tech360 before activating the ONU and replacing the gateway.

1. Add the ONU device in Tech360.
2. Swap the DOCSIS gateway for the Technicolor XB7 or Sercomm XER10 router.
3. Activate services in Tech360.
4. Ensure that WiFi settings and all other services are transferred and working as expected.

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